# Predicting NFL Margin of Victory: *Machine Learning vs. Vegas*

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# Refresher

# Recap

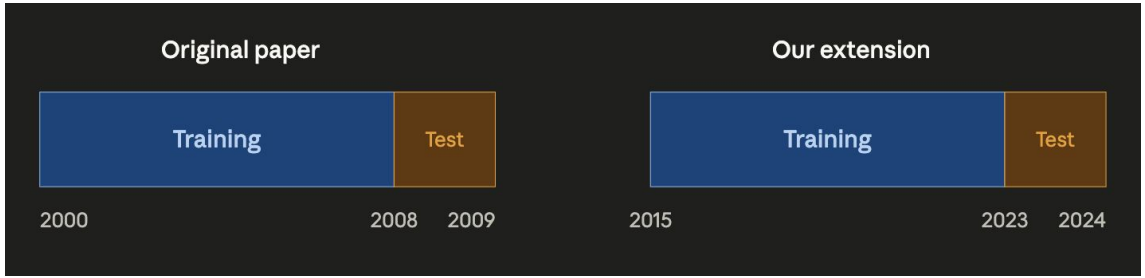
- **Research Question:** Is it possible to create a machine learning framework utilizing Gaussian Processes to predict margin of victory for any given NFL matchup, specifically with the goal of a win rate that is at least 52.4%?
- For each game, the features included are (all of which, besides computed strength, have data for the home team, away team, home team streak, and away team streak):
    - Winning %
    - Points Per Game Scored, Points Per Game Allowed
    - Total Yards Per Game Gained, Total Yards Per Game Allowed
    - Pass Yards Per Game Gained, Pass Yards Per Game Allowed
    - Rush Yards Per Game Gained, Rush Yards Per Game Allowed
    - Turnovers taken per game, Turnovers Given up per game
    - Computed Strength
- **Key Finding**: Vegas point spreads slightly outperformed GP margin predictions and mean absolute error between the two was roughly the same → no edge for profitability

# Our Extension

- Preserving the same methodology, kernel, and feature structure as the paper, we want to answer the core question:

  **Has the Vegas spread become “sharper” and gotten harder to predict over the past 15 years?**

- Shifting our data window to modernize the paper



# GP Overview

# What is a Gaussian Process?

- It is a non-parametric Bayesian machine learning method that represents a distribution over functions
- Intuition is that similar inputs should produce similar outputs (i.e. two games with similar team stats should have similar predicted margins)
- The kernel is how we measure “similar”

# GP vs. Linear Regression

| | **Gaussian Process** | **Linear Regression** |
|---|---|---|
| Model | Non-parametric, highly flexible | Parametric |
| Output | Mean + variance | Point estimate |
| Uncertainty | Grows where data is sparse | Same everywhere |
| Computational Complexity | O(N^3) | O(N) |
| Use Case | Precision/Small Data | Speed/Interpretability |

# The Kernel

- RBF (Squared Exponential) Kernel Formula:

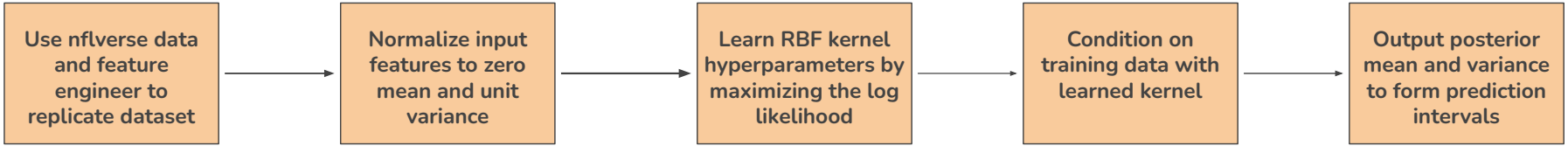
$$k(\vec{x}, \vec{x}') = \sigma_f^2 \exp\left(-\frac{1}{2}(\vec{x} - \vec{x}')^T M (\vec{x} - \vec{x}')\right)$$

  - $\sigma_f^2$ = signal variance (large values with big swings vs. small, tight variations)
  - **M** controls how much each feature contributes to distance
    - Smaller the $l_i$ → feature i matters more (inverse-squared in diagonal matrix)
- White kernel added on top to capture irreducible noise $\sigma_n^2$ (injuries, weather, randomness) between training data

$$\begin{bmatrix} y \\ f^* \end{bmatrix} \sim N\left(0, \begin{bmatrix} K(X,X) + \sigma_n^2 I & K(X,X^*) \\ K(X^*,X) & K(X^*,X^*) \end{bmatrix}\right)$$

- The prior is encoded with the kernel → can combine different types of kernels for complex patterns

# Our GP Model Pipeline



# Feature Selection

- **Cross Validation feature selection scheme**
  - Fix home and away win percentage as base features (most indicative of the outcome)
  - Add remaining features one-at-a-time with cross-validation
  - Keep top 20 lowest-CV-error features as search set
  - Run forward selection, adding only error-reducing features, on training data
  - Stop at four features because additional inputs increase error

| | Feature | Error |
|---|---|---|
| 1 | Computed Strength (A) | 10.913 |
| 2 | Pass yards per game allowed (H/S) | 10.937 |
| 3 | Points per game scored (A) | 10.949 |
| 4 | Total yards per game gained (H/S) | 10.950 |
| 5 | Computed Strength (H) | 10.951 |
| 6 | Total yards per game allowed (H) | 10.969 |
| 7 | Points per game scored (A/S) | 10.979 |
| 8 | Pass yards per game gained (A/S) | 10.982 |
| 9 | Pass yards per game gained (H/S) | 11.000 |
| 10 | Rush yards per game allowed (A/S) | 11.006 |
| ... | ... | ... |
| 39 | Temperature Difference | 11.161 |

Top 10 performing features

| Final Feature Set #1 |
|---|
| Winning Percentage (H) |
| Winning Percentage (A) |
| Computed Strength (A) |
| Pass yards per game gained (A/S) |

Final feature set (only four features!)

# Results

# Original model, modern data

| | GP (Warner 2008-09) | Vegas (Warner 2008-09) | GP (Ours 2023-24) | Vegas (Ours 2023-24) |
|---|---|---|---|---|
| MAE (margin of victory) | ~11.5 | ~11.3 | 9.98 | **9.39** |
| Winner accuracy | 64.4% | **68.5%** | 64.4% | **72.1%** |

- Using the same four features as paper: Same model, new data
  - (H/A win %, Away Keener strength, Away Pass yds streak)
- Mean Absolute Error improves, but this doesn't mean the model has improved
  - Winner Accuracy remained the same
  - This likely just reflects the league becoming more competitive
- **Main takeaway:** Vegas has gotten sharper!

# Reselect Features

- Home win % (season avg)
  *H_win_pct*
- Away win % (season avg)
  *A_win_pct*
- Away points scored (season avg)
  *A_points_scored_avg*
- Home Keener (eigenvalue) strength
  *H_computed_strength*
- Away total yards (season avg)
  *A_total_yards_avg*
- Away total yards (4-game streak)
  *A_total_yards_streak*
- Home points allowed (4-game streak)
  *H_points_allowed_streak*
- Away turnovers lost (season avg)
  *A_turnovers_lost_avg*

- Re-ran Warner's exact search on our training window to find what's actually predictive now
- Two stages: screen all 44 candidates by CV MAE, then greedily forward-search the top 20
- Search stopped at 8 features, 5 of them away-team stats
- For every candidate and for every CV fold in the forward search, you train a new GP
  - You can imagine how long this took to run!

# Different features, modern data

| | Warner 4-Feature GP (2023-24 test) | Our 8-Feature GP (2023-24 test) | Vegas (2023-24 test) |
|---|---|---|---|
| MAE (margin of victory) | 9.98 | 10.02 | **9.39** |
| Winner accuracy | 64.4% | **67.1%** | **72.1%** |

- Now we use a GP on modern data with the 8 newly selected features
- Mean Absolute Error improves in the same way as the 4-feature GP, but
  - Winner Accuracy makes significant jump
  - With better calibrated features, the GP is a stronger predictor
    - But not in terms of MAE
- Still, pales in comparison to Vegas

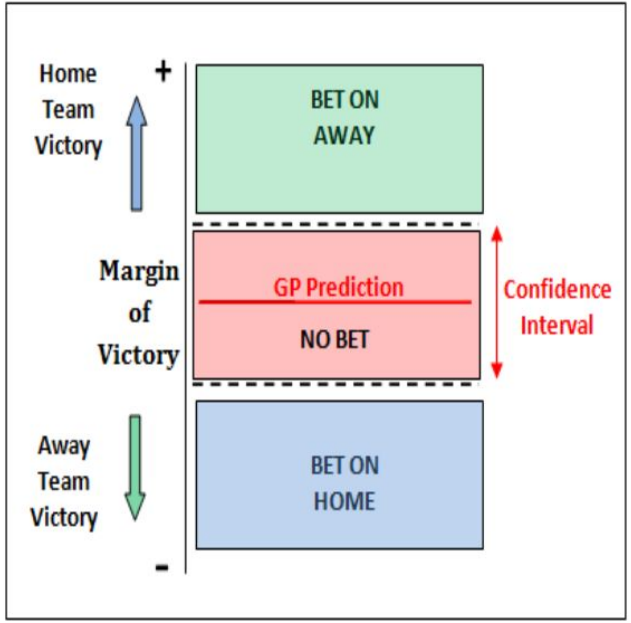
# Betting Scheme

The paper's betting scheme:

- Compare Vegas spread to GP 95% confidence interval
- Bet home if spread below interval; bet away if above
- Place no bet when spread lies within interval

What we added: k-values

- Instead of a fixed 95% interval, vary the threshold k
  - Bet only when the spread differs from the GP prediction by more than k standard deviations
- Higher k = fewer but higher-confidence bets



# Betting Results

## 4-Feature GP (2023-2024)

| | Total Bets | Wins | Win Rate | Profitable? |
|---|---|---|---|---|
| k = 0.25 | 158 | 75 | 47.5% | no |
| k = 0.50 | 39 | 20 | 51.3% | no |
| k = 0.75 | 5 | 4 | 80.0% | YES |
| k = 1.00 | 1 | 0 | 0.0% | no |
| k = 1.25 | 0 | 0 | — | no |
| k = 1.50 | 0 | 0 | — | no |

## 8-Feature GP (2023-2024)

| | Total Bets | Wins | Win Rate | Profitable? |
|---|---|---|---|---|
| k = 0.25 | 159 | 77 | 48.4% | no |
| k = 0.50 | 37 | 18 | 48.6% | no |
| k = 0.75 | 12 | 7 | 58.3% | YES |
| k = 1.00 | 1 | 1 | 100.0% | YES |
| k = 1.25 | 0 | 0 | — | no |
| k = 1.50 | 0 | 0 | — | no |

- Sweep k from 0.25 to 1.5 to find if any threshold produces a profitable win rate (52.4%)
- But wait - Warner was producing bets with a 95% interval, k = 1.96...
  - At k = 1.96, the spread would need to differ from the GP by over 26 points
  - Why did this happen more often for 2008-2009? *Vegas was worse*
  - So accurate that the GP almost never sees a spread it disagrees with enough

# Could it be the GPs fault?

## 8-Feature GP (2023-2024)

| | Total Bets | Wins | Win Rate | Profitable? |
|---|---|---|---|---|
| k = 0.25 | 159 | 77 | 48.4% | no |
| k = 0.50 | 37 | 18 | 48.6% | no |
| k = 0.75 | 12 | 7 | 58.3% | YES |
| k = 1.00 | 1 | 1 | 100.0% | YES |
| k = 1.25 | 0 | 0 | — | no |
| k = 1.50 | 0 | 0 | — | no |

## 8-Feature Random Forest (2023-2024)

| | Total Bets | Wins | Win Rate | Profitable? |
|---|---|---|---|---|
| k = 0.25 | 264 | 135 | 51.1% | no |
| k = 0.50 | 135 | 65 | 48.1% | no |
| k = 0.75 | 49 | 25 | 51.0% | no |
| k = 1.00 | 20 | 10 | 50.0% | no |
| k = 1.25 | 6 | 3 | 50.0% | no |
| k = 1.50 | 3 | 2 | 66.7% | YES |

- Random Forest is a known strong baseline for tabular data
- RF places more bets at every k level because its tree-variance uncertainty is smaller and more calibrated than the GP's posterior σ
- Neither model finds a consistent profitable edge
- Consistent failure across two very different model architectures

# Takeaways

# Main Takeaways

- **Primary takeaway:** Vegas has gotten significantly sharper since 2010
  - Winner accuracy jumped from 68.5% to 72.1% while the GP's stayed flat at ~64%
- Box-score stats alone are not close to enough to compete with the modern sportsbook
- Feature selection helped pick winners (64.4% → 67.1%) but couldn't close the gap
- Random Forest also fail to find a profitable betting edge
  - Evidence that the model isn’t the problem
- Warner's 2010 conclusion was that there may exist a small, but unprofitable, edge
  - Now, even that small potential edge has evaporated
- More careful data selection and advanced analytics are required to compete

# Limitations

- That’s not to say Warner’s model is the best you can do…
- No injury data
  - Star QB missing a game is invisible to the model but immediately priced into the Vegas line
- No weather or stadium conditions
- No line movement
- Forward search is greedy and not guaranteed to find the optimal feature set
  - Different combination of features might do better
- So many variables Vegas considers, some we can’t even conceive… A basic model like this never had a chance

# Assignment

# Jupyter Notebook

- The notebook takes you through data exploration, building a GP, and evaluating it versus vegas
- No coding – just interpreting